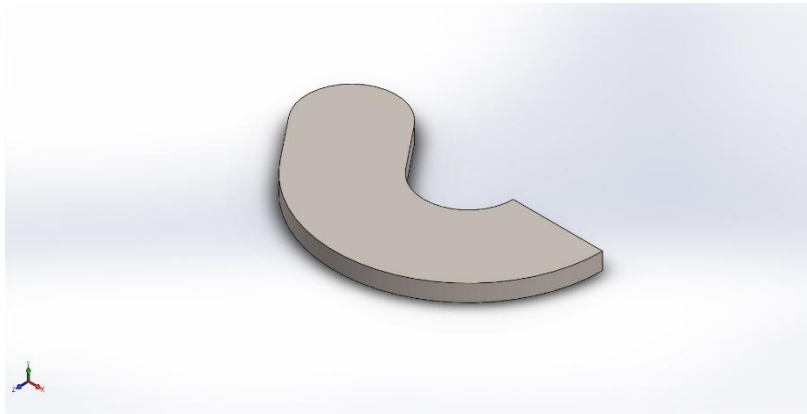




Description

No Data

Simulation of CraneHook_75in_x_4in_SectionAA_Study

Date: Saturday, January 14, 2023

Designer: MD Naiem Gazi

Study name: Hook

Analysis type: Static

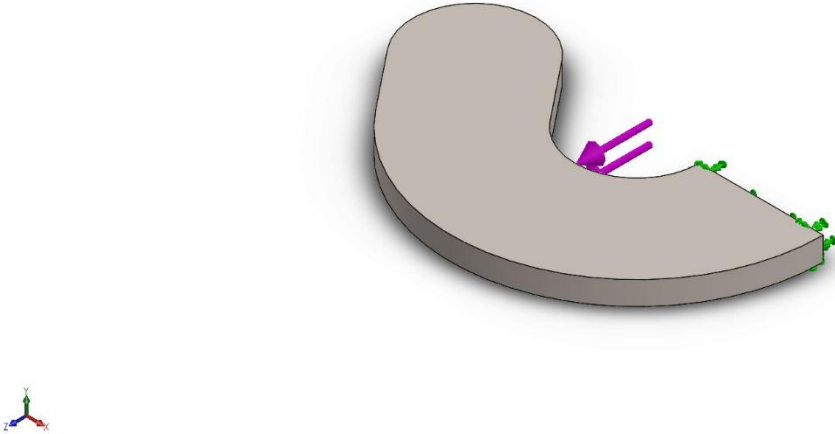
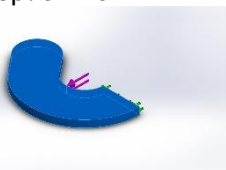
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Assumptions

Model Information

 Model name: Crane_Hook Current Configuration: Default			
Solid Bodies			
Document Name and Reference	Treated As	Volumetric Properties	Document Path/Date Modified
Split Line1 	Solid Body	Mass:5.26117 kg Volume:0.000683268 m ³ Density:7,700 kg/m ³ Weight:51.5594 N	E:\L-3 T- 2\Assignments\part 4\Part4.SLDPRT Jan 13 23:22:54 2023

Study Properties

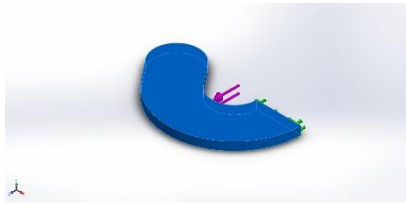
Study name	CraneHook_75in_x_4in_SectionAA_Study
Analysis type	Static
Mesh type	Solid Mesh
Thermal Effect:	On
Thermal option	Include temperature loads
Zero strain temperature	298 Kelvin
Include fluid pressure effects from SOLIDWORKS Flow Simulation	Off
Solver type	Automatic
Inplane Effect:	Off
Soft Spring:	Off
Inertial Relief:	Off
Incompatible bonding options	Automatic
Large displacement	Off
Compute free body forces	On
Friction	Off
Use Adaptive Method:	Off
Result folder	SOLIDWORKS document (E:\L-3 T-2\Assignments\part 4)

Units

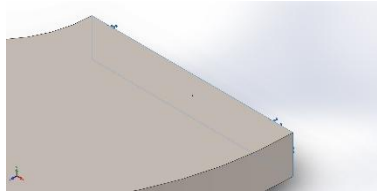
Unit system:	SI (MKS)
Length/Displacement	mm
Temperature	Kelvin
Angular velocity	Rad/sec
Pressure/Stress	N/m ²

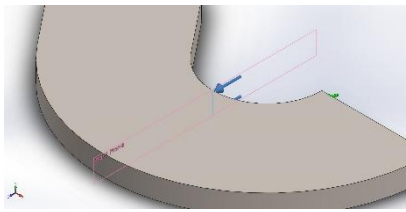


Material Properties

Model Reference	Properties	Components
	Name: Alloy Steel Model type: Linear Elastic Isotropic Default failure criterion: Max von Mises Stress Yield strength: 6.20422e+08 N/m ² Tensile strength: 7.23826e+08 N/m ² Elastic modulus: 2.1e+11 N/m ² Poisson's ratio: 0.28 Mass density: 7,700 kg/m ³ Shear modulus: 7.9e+10 N/m ² Thermal expansion coefficient: 1.3e-05 /Kelvin	SolidBody 1(Split Line1)(Part4)
Curve Data:N/A		

Loads and Fixtures

Fixture name	Fixture Image	Fixture Details		
Fixed-1		Entities: 1 face(s) Type: Fixed Geometry		
Resultant Forces				
Components	X	Y	Z	Resultant
Reaction force(N)	-0.250536	0.993803	-22,235.1	22,235.1
Reaction Moment(N.m)	0	0	0	0

Load name	Load Image	Load Details
Force-1		Entities: 1 edge(s), 1 plane(s) Reference: Right Plane Type: Apply force Values: -5,000, ---, --- lbf

Connector Definitions

No Data

Contact Information

No Data

Mesh information

Mesh type	Solid Mesh
Mesher Used:	Blended curvature-based mesh
Jacobian points for High quality mesh	16 Points
Maximum element size	0.173423 in
Minimum element size	0.173423 in
Mesh Quality	High

Mesh information - Details

Total Nodes	108180
Total Elements	72667
Maximum Aspect Ratio	3.6883
% of elements with Aspect Ratio < 3	99.9
Percentage of elements with Aspect Ratio > 10	0
Percentage of distorted elements	0
Time to complete mesh(hh:mm:ss):	00:00:05
Computer name:	HELAL



Sensor Details

No Data

Resultant Forces

Reaction forces

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N	-0.250536	0.993803	-22,235.1	22,235.1

Reaction Moments

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N.m	0	0	0	0

Free body forces

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N	3.46496	13.1824	4.5187	14.3597

Free body moments

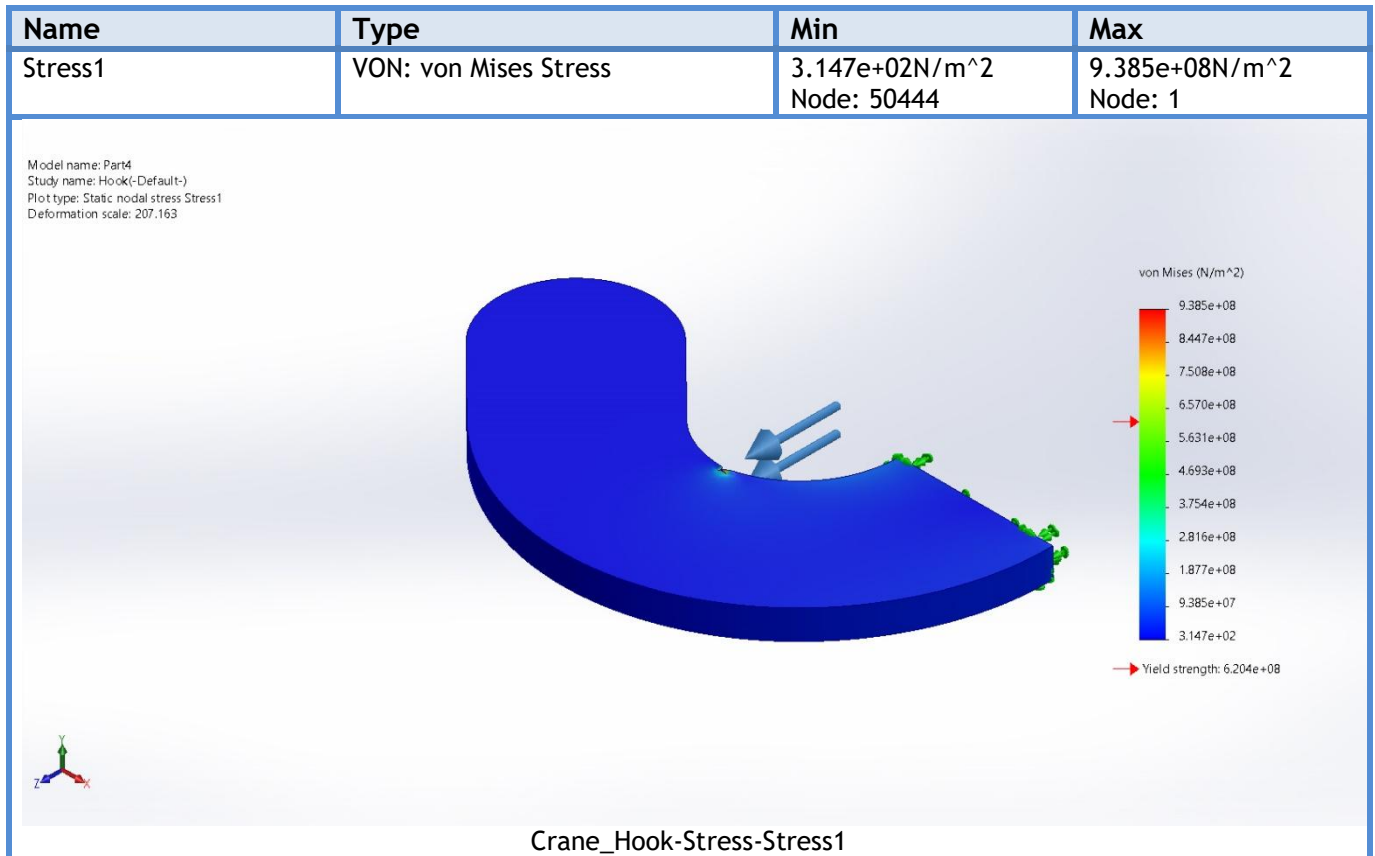
Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N.m	0	0	0	1e-33

Beams

No Data

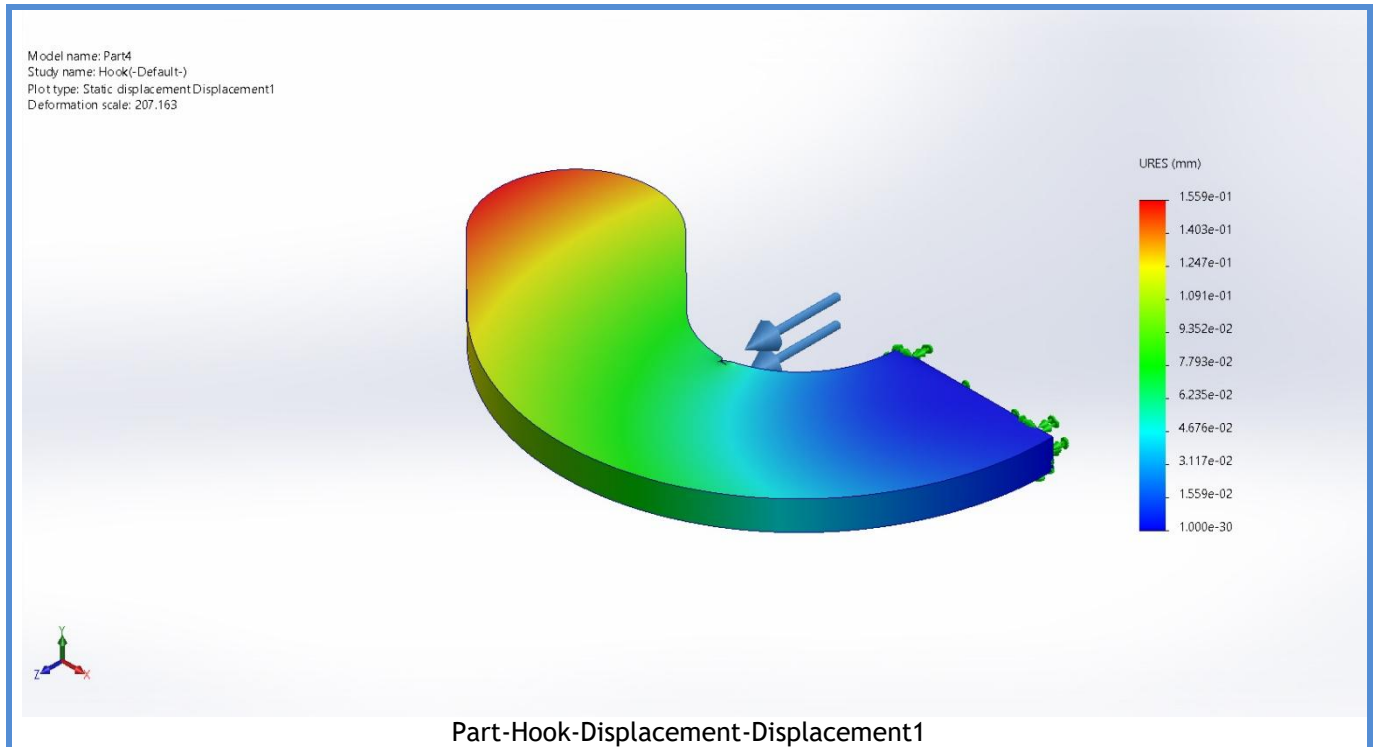


Study Results

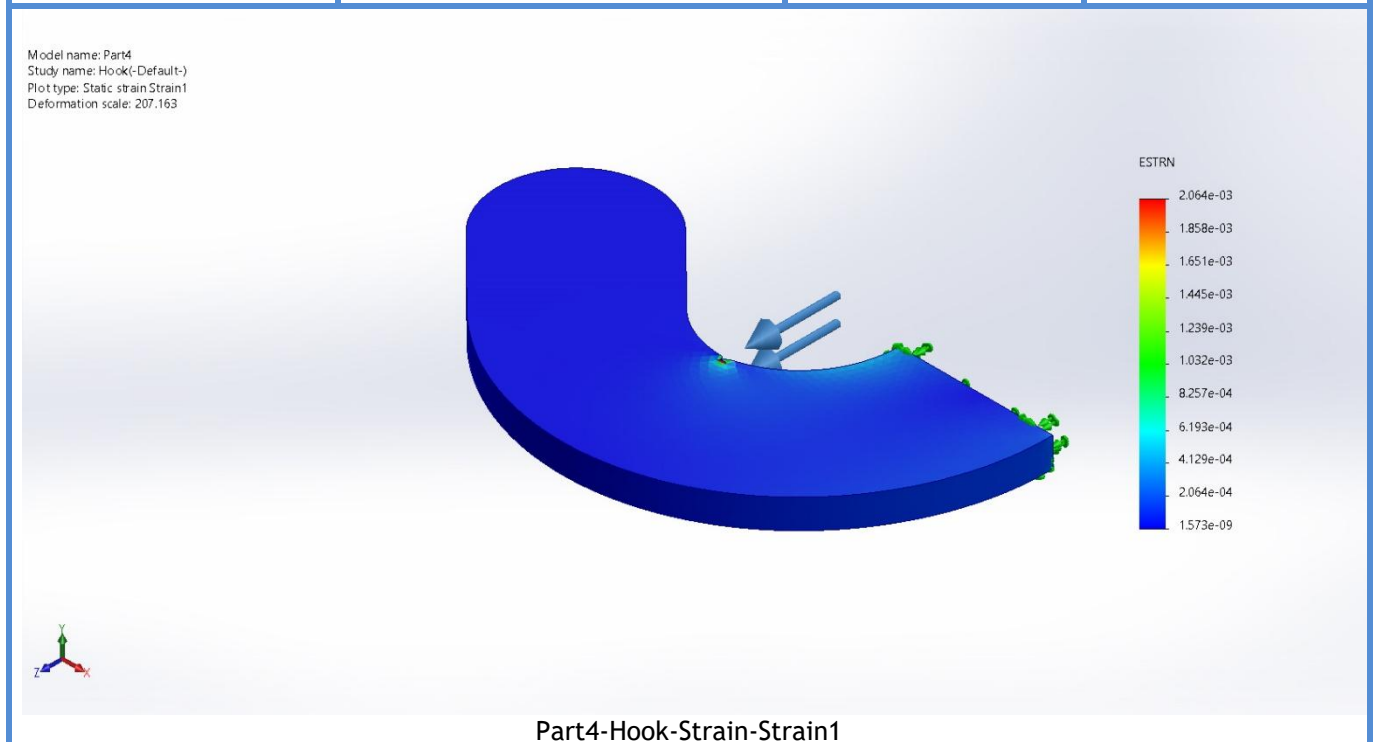


Name	Type	Min	Max
Displacement1	URES: Resultant Displacement	0.000e+00mm Node: 6	1.559e-01mm Node: 387





Name	Type	Min	Max
Strain1	ESTRN: Equivalent Strain	1.573e-09 Element: 5322	2.064e-03 Element: 37684



Conclusion

